

CIT304: Tutorial 1

Question 1

Problem Statement

FinTrust Digital Bank is a fast-growing B2C digital banking company that offers personal loans, credit cards, savings accounts, and Buy-Now-Pay-Later (BNPL) services through a mobile app.

In the last two years, customer growth increased by 40%. However, the bank is now facing serious performance issues:

- Non-Performing Loan (NPL) ratio increased from 2.1% to 4.8%
- Customer Acquisition Cost (CAC) increased by 35%
- Online fraud attempts have doubled
- The cost-to-income ratio is rising
- No single dashboard exists for executives to monitor overall performance

Data is currently scattered across multiple systems such as core banking, CRM, fraud monitoring tools, mobile app analytics, and marketing platforms. The CEO wants to implement a Business Intelligence (BI) system to integrate data, improve risk control, reduce costs, monitor fraud in real time, and support better decision-making at all levels of the organization.

Question

As a BI consultant, propose a practical Business Intelligence solution for FinTrust Digital Bank to improve profitability, reduce credit risk, control fraud, and support better strategic and operational decisions.

1. Briefly explain how you would design the BI architecture.

- What main data sources should be integrated?
- What type of analytics models should be used?

2. What key KPIs should be shown to the CEO and board? How would this dashboard support long-term strategic decisions?

3. How can Credit Risk, Marketing, and Fraud Managers use BI dashboards? What reports or insights would help them improve performance?

4. How can frontline staff (loan officers, fraud analysts, customer service agents) use BI tools in their daily work? What kind of real-time alerts or reports are needed?

5. KPI Framework & Business Model

a) Group the KPIs under:

- Profitability
- Risk
- Efficiency
- Growth
- Customer Value

b) Identify FinTrust's business model and explain how BI supports long-term competitive advantage.